

Requirements Specification

What the Registrar Management Information System is required to do, what it is required to be, and — stated honestly against each clause — how much of it the current build actually does.

Version: V9.4 **Status date:** 9 September 2026 **Scope:** front-end prototype, in-browser simulated server

CONVENTIONS

How to read this document

Every clause carries an identifier and a conformance mark. The marks are not aspirational — each was checked against the code before it was written down, and a clause marked *Not built* is one no screen currently satisfies.

- IMPLEMENTED

the build satisfies the clause in full
- PARTIAL

satisfied in part; the shortfall is named in the clause
- NOT BUILT

no implementation exists — carried here so it is not mistaken for done

PART ONE

Functional requirements

FR-01 Authentication & access control

FR-01.1	The system shall authenticate users by centre email address and password, and shall lock an account for 15 minutes after five consecutive failed attempts.	IMPLEMENTED
FR-01.2	The system shall support four roles — Registrar, Trainer, Trainee, Admin/IT — and shall present each user only the modules their role permits.	IMPLEMENTED
FR-01.3	Role restrictions shall be enforced on the server side of the API boundary, independently of what the interface displays. Enforced by <code>requireRole()</code> in every service. Hiding a control is treated as a courtesy, never as the control itself.	IMPLEMENTED

FR-01.4	The system shall provide an interface for creating, editing and deactivating user accounts. The account-administration screen was removed in V9 when the application was narrowed. Roles and sign-in remain; management of accounts does not.	NOT BUILT
---------	---	-----------

FR-02 Public admission application

FR-02.1	An applicant shall be able to submit a registration form without an account, encoding their own personal, contact, address, educational and emergency-contact details.	IMPLEMENTED
FR-02.2	The emergency contact shall be recorded as last name, first name and middle name separately, not as a single combined field.	IMPLEMENTED
FR-02.3	The applicant shall upload an identification photograph and a birth certificate, and shall be shown the required filename before uploading — built from their own entered name, e.g. <code>CRUZ_MARIA_ID</code> .	IMPLEMENTED
FR-02.4	A submitted application shall appear in the Registrar's Pending list without manual re-entry, and shall issue the applicant a reference code.	IMPLEMENTED
FR-02.5	The system shall refuse a duplicate application. An applicant matching an existing non-rejected record by full name, or by email address, shall be turned away and directed to the Registrar. A previously rejected applicant may reapply — refusing them on the basis of the refusal would be wrong.	IMPLEMENTED

FR-03 Trainee records

FR-03.1	The system shall hold one centralised profile per trainee covering personal information, contact and address, educational background, admission basis and Diploma.	IMPLEMENTED
FR-03.2	The Registrar shall review, correct and approve or reject a submitted application, and approval shall assign the trainee's curriculum and section.	IMPLEMENTED
FR-03.3	A trainee shall carry one of the statuses <code>PENDING</code> , <code>APPROVED</code> , <code>REJECTED</code> , <code>ACTIVE</code> , <code>INACTIVE</code> , <code>GRADUATED</code> or <code>DROPPED</code> .	IMPLEMENTED

FR-03.4	The system shall record and display the date and time an application was submitted and the date and time it was approved, with a relative label such as <i>5 mins ago</i> .	IMPLEMENTED
FR-03.5	The Registrar shall be able to archive a trainee record, and archiving shall require the Registrar to re-enter their own password.	IMPLEMENTED
FR-03.6	The Registrar shall be able to import trainee records in bulk from a CSV file, with every invalid row reported in a single pass.	IMPLEMENTED
FR-04 Academic catalogue & calendar		
FR-04.1	The system shall maintain Diplomas, curricula, subjects and sections, and shall support mapping a subject into a curriculum at a stated year level and semester.	IMPLEMENTED
FR-04.2	A semester shall belong to a Diploma, year level and period rather than to the centre as a whole, so that Diplomas may run independent calendars.	IMPLEMENTED
FR-04.3	Exactly one semester shall be open per Diploma and year level at any time; opening one shall close the others.	IMPLEMENTED
FR-04.4	A subject mapping shall carry its prerequisite subjects, any year-standing requirement, and the curriculum's own prerequisite wording for printing.	IMPLEMENTED
FR-05 Class scheduling		
FR-05.1	The Registrar shall create class schedules assigning a subject to a section, a trainer, a room, days and times within a semester.	IMPLEMENTED
FR-05.2	The system shall detect and refuse scheduling conflicts for a room, a section or a trainer.	IMPLEMENTED
FR-05.3	A schedule shall be held as a draft until published, and only published classes shall be available for enrolment.	IMPLEMENTED
FR-05.4	A trainer shall view their own weekly teaching timetable, scoped to their own published classes.	IMPLEMENTED

FR-06 Enrolment

FR-06.1	The system shall load the subjects available to a trainee from their own curriculum for the selected year level and semester.	IMPLEMENTED
FR-06.2	Once a trainee is selected, the Diploma shall be fixed to the one they were approved for, and no other Diploma shall be offered.	IMPLEMENTED
FR-06.3	A trainee shall hold one enrolment per semester ; a second attempt shall be refused by name and term.	IMPLEMENTED
FR-06.4	Sequential Enrolment. A trainee proceeding to their next semester in sequence shall be refused until every grade in the preceding semester is on record, naming the outstanding subjects. The Registrar may override with a stated reason.	IMPLEMENTED
FR-06.5	Prerequisites shall be enforced at enrolment, not merely displayed. A subject shall be refused where a prerequisite is failed, incomplete, unresolved, still in progress, or never taken. Every unsatisfied prerequisite shall be named, not only the first.	IMPLEMENTED
FR-06.6	The Registrar shall drop an individual subject from an enrolment, with a stated reason, provided the subject does not yet carry a grade.	IMPLEMENTED
FR-06.7	Units shall be copied onto the enrolment at the moment of enrolling and never re-read from the subject afterwards.	IMPLEMENTED
FR-06.8	The system shall generate a printable or exportable enrolment list or report. An on-screen list of the term's enrolments exists. There is no export or print output.	PARTIAL

FR-07 Grading sheets

FR-07.1	A trainer shall encode grades for their own classes on the 1.00–5.00 scale, or record the markers INC , DRP or NG .	IMPLEMENTED
FR-07.2	Submitting a sheet shall route it to the Registrar for review and shall issue a reference number by which both parties may refer to it.	IMPLEMENTED
FR-07.3	The Registrar shall approve a sheet or send it back to the trainer with a stated reason. A returned sheet shall reopen pre-filled, so the trainer edits rather than re-keys.	IMPLEMENTED

FR-07.4	Approval shall be the only path by which a grade reaches a trainee's record. A submitted sheet shall post nothing.	IMPLEMENTED
FR-07.5	Where a trainee joins a class after its sheet has been reviewed, the sheet shall reopen with the trainee added ungraded and a remark naming them. Grades already entered and already posted shall be preserved.	IMPLEMENTED
FR-07.6	The review queue shall group sheets by Diploma, allow each group to be folded, and sort on any column.	IMPLEMENTED
FR-07.7	In addition to the per-subject view, the Registrar shall open a student-level grading sheet aggregating one trainee's grades across all semesters, reachable from the Submitted, Pending and Approved tabs.	IMPLEMENTED
FR-08 Grade evaluation & INC resolution		
FR-08.1	The system shall compile a Grade Evaluation Form covering every semester on record, with subject, units, prerequisite, grade, remarks and a unit-weighted GWA.	IMPLEMENTED
FR-08.2	The form shall be produced whether or not all grades are in, and shall state how many subjects remain ungraded rather than leaving blanks to be misread.	IMPLEMENTED
FR-08.3	An unresolved INC shall force the GWA to 0.000 as a deliberate signal that the average cannot yet be relied upon.	IMPLEMENTED
FR-08.4	The Registrar shall resolve an INC either as a completion — the work was finished, and the INC remains visible on the record beside its completion grade — or as a correction , where the INC was recorded in error. Both shall require remarks.	IMPLEMENTED
FR-08.5	The form shall print alone, without the surrounding interface.	IMPLEMENTED
FR-09 General Schedule & Assessment		
FR-09.1	The system shall produce a trainee's weekly schedule and course load for their currently open semester, ready to print.	IMPLEMENTED

FR-10 Trainee portal

FR-10.1	A trainee shall view their own weekly schedule, their own grades and their own student identification, and shall not be able to alter any of them.	IMPLEMENTED
---------	--	-------------

FR-11 Audit trail

FR-11.1	Every action that changes a record shall be written to an audit log recording who acted, what changed, when, and — where a reason is required — why.	IMPLEMENTED
---------	--	-------------

FR-11.2	The audit log shall be searchable and filterable by action and record type.	IMPLEMENTED
---------	---	-------------

FR-12 Academic progress & status tracking

FR-12.1	The system shall present a trainee's subject-completion progression across semesters. Satisfied by the Grade Evaluation Form.	IMPLEMENTED
---------	--	-------------

FR-12.2	The system shall flag retention, withdrawal and transfer statuses distinctly and report on them. Statuses exist for dropped, inactive and graduated, and a transferee flag exists on intake. There is no retention or transfer tracking, and no reporting on any of them.	PARTIAL
---------	--	---------

FR-13 Completion & graduation

FR-13.1	The system shall determine which trainees have met completion requirements and flag them automatically. Fields for graduation date and special order number exist on the record but are never computed or set. No eligibility rule exists.	NOT BUILT
---------	---	-----------

FR-13.2	The system shall generate printable completion and graduation lists.	NOT BUILT
---------	--	-----------

FR-14 Certification & document requests

FR-14.1	The system shall log a trainee's request for a Transcript of Records, Certificate of Enrolment or other Registrar document, and track it through to issuance. A data shape for document requests is declared but nothing uses it. This is distinct from the admission-requirement uploads, which are documents coming in rather than requests going out.	NOT BUILT
---------	---	-----------

FR-14.2	The system shall generate a Transcript of Records.	NOT BUILT
	No transcript generator exists. The Grade Evaluation Form is presently the cumulative academic record.	

FR-15 Reports & statistics

FR-15.1	The Dashboard shall present live counts of trainees, enrolments, pending applications and published schedules.	IMPLEMENTED
FR-15.2	The system shall generate automated enrolment, academic, completion and statistical reports for the Registrar and management.	NOT BUILT
	No reporting module exists. The Dashboard's counts are the whole of the current reporting capability.	

PART TWO

Non-functional requirements

NFR-01 Data integrity

NFR-01.1	Every operation shall validate completely before writing anything. No operation shall apply in part.	IMPLEMENTED
NFR-01.2	The grading scale shall be defined in exactly one place and reused, so that the transcript, the weighted average, the evaluation form and the prerequisite check cannot disagree about what constitutes a pass.	IMPLEMENTED
	1.00 highest, 5.00 lowest, 3.00 the passing cutoff and the 75% equivalent. There is no percentage layer anywhere in the system.	
NFR-01.3	The system shall not permit two records to describe the same person.	IMPLEMENTED
NFR-01.4	Multi-write operations shall be atomic.	PARTIAL
	Presently guaranteed by the store being synchronous and single-threaded rather than by transactions. A real database will require explicit transactions — approving a grading sheet posts many grades and changes a status, and must not half-apply.	

NFR-02 Security & privacy

NFR-02.1	Authorisation shall be enforced behind the API boundary, so that a request bypassing the interface is refused on the same grounds and in the same words.	IMPLEMENTED
----------	--	-------------

NFR-02.2	Destructive actions shall require the acting user to re-enter their own password.	IMPLEMENTED
NFR-02.3	Every screen shall carry the confidentiality notice required under the Data Privacy Act of 2012 (RA 10173).	IMPLEMENTED
NFR-02.4	Passwords shall be stored hashed, never in plain text. Demonstration accounts are seeded in plain text. This is acceptable only because no real credential exists in the system, and must be resolved before any real deployment.	NOT BUILT
NFR-03 Maintainability		
NFR-03.1	The interface shall never reach the data store directly. All access shall pass through a typed client and a single server surface.	IMPLEMENTED
NFR-03.2	The transport layer shall be replaceable without altering any screen, so that a real backend can be introduced behind the existing boundary.	IMPLEMENTED
NFR-03.3	The project shall compile under TypeScript strict mode with no errors.	IMPLEMENTED
NFR-04 Usability & accessibility		
NFR-04.1	Navigation shall be scoped to role, so no user is shown a destination they cannot enter.	IMPLEMENTED
NFR-04.2	Refusals shall explain what is wrong and what to do about it, naming the record or grade concerned.	IMPLEMENTED
NFR-04.3	Sortable and collapsible controls shall be operable by keyboard and shall expose their state to assistive technology.	IMPLEMENTED
NFR-04.4	The interface shall render legibly in light and dark themes and at tablet and phone widths.	IMPLEMENTED
NFR-04.5	Printed output shall contain the document alone, with no surrounding interface.	IMPLEMENTED

NFR-05 Performance

NFR-05.1	Ordinary operations shall complete within one second. All data is held in memory; a deliberate 120 ms delay is added so the interface behaves as it would against a network.	IMPLEMENTED
----------	---	-------------

NFR-06 Portability & deployment

NFR-06.1	The built application shall run on any static host without configuration change. Relative asset paths and fragment-based routing; the same build runs on GitHub Pages, Netlify, Vercel or Cloudflare Pages unmodified.	IMPLEMENTED
----------	---	-------------

NFR-06.2	Deployment shall be automatic on publication to the main branch.	IMPLEMENTED
----------	--	-------------

NFR-07 Persistence & availability

NFR-07.1	Records shall survive a page reload.	IMPLEMENTED
----------	--------------------------------------	-------------

NFR-07.2	Records shall be shared between users and devices. The decisive limitation of the present build. Data is held in each browser, so two machines never see the same records. Tabs within one browser do stay in step. Resolving this requires a database and a server runtime.	NOT BUILT
----------	---	-----------

NFR-07.3	The system shall take automated backups and support a tested restore.	NOT BUILT
----------	---	-----------

NFR-08 Demonstration integrity

NFR-08.1	The system shall identify itself as a demonstration prototype, unaffiliated with TESDA, on every page reachable without signing in — in the page title, above the fold, and in the footer. Required, not cosmetic: a page carrying a government agency's name and seal, requesting a password and collecting birth certificates, was classified as a deceptive site until this disclosure was added.	IMPLEMENTED
----------	---	-------------

NFR-08.2	The Registrar shall restore the prepared demonstration dataset on demand, behind a typed confirmation and a password.	IMPLEMENTED
----------	---	-------------

Conformance summary

AREA	IMPLEMENTED	PARTIAL	NOT BUILT
Functional (FR-01 – FR-15)	36	2	5
Non-functional (NFR-01 – NFR-08)	18	2	3

The five functional gaps, in the order they should be closed

- **Reports & statistics (FR-15.2)** — the largest visible gap, and the first thing a Registrar will ask for.
- **Completion & graduation (FR-13)** — without it the trainee workflow has no ending.
- **Document requests (FR-14.1)** — self-contained; a data shape already exists to build against.
- **Transcript of Records (FR-14.2)** — decide first whether it supersedes the Grade Evaluation Form or sits beside it.
- **User account administration (FR-01.4)** — only if the deployed system is to manage its own accounts.

What the build has that the original plan did not ask for

Sequential Enrolment with enforced prerequisites · class scheduling with conflict detection · the General Schedule and Assessment · the public application form with document filing · restorable demonstration data · a full audit trail across every module.